

1	ACSB/2025/I/02
	(a) Factorise $x^3 - y^3$. [2] (b) Hence express $\frac{7^{\frac{3}{2}} - 2^{\frac{3}{2}}}{\sqrt{7} - \sqrt{2}}$ in the form $a + b\sqrt{14}$, where a and b are integers. [3]
2	ACSB/2025/I/05
	Express $\frac{2x^3 - 2x^2 - 24x - 7}{x^2 - x - 12}$ in partial fractions. [5]
3	ACSB/2025/II/04
	It is given that $f(x) = x^3 + mx^2 + 6x - 4$, where m is a constant. (a) Show that $m = 6$ given that $x + 2$ is a factor of $f(x)$. [2] (b) Using the value of m in part (a), solve the equation $f(x) = 0$, expressing non-integer roots in surd form. [4]
4	BASS/2025/I/03
	(a) Explain why $\frac{x^3 + 5x^2 - 14x - 10}{x^2 + 4x - 21}$ is not a proper fraction. [1] (b) Express $\frac{x^3 + 5x^2 - 14x - 10}{x^2 + 4x - 21}$ in partial fraction. [6]
5	BASS/2025/II/02
	The remainder when the expression $2hx^3 + kx^2 + 4x + 4$ is divided by $(x - 2)$ is four times the remainder when it is divided by $(x + 1)$. (a) Given that $(x - 1)$ is a factor of the expression, find the value of h and of k. [4] (b) Hence solve $2hx^3 + kx^2 + 4x + 4 = 0$, leaving your answers in the form $a \pm b\sqrt{3}$ where applicable. [4]
6	GMS/2025/II/10
	The cubic polynomial $f(x)$ is such that the coefficient of x^3 is -1. The roots of the equation $f(x) = 0$ are 1, 2 and k. Given that $f(x)$ has a remainder of 8 when divided by $(x - 3)$, find (a) the value of k, [3] (b) the remainder when $f(x)$ is divided by $(x + 3)$. [1]

1	(a) $(x-y)(x^2+xy+y^2)$ (b) $\frac{(\sqrt{7})^3 - (\sqrt{2})^3}{\sqrt{7} - \sqrt{2}}$$= \frac{(\sqrt{7} - \sqrt{2})(7 + \sqrt{14} + 2)}{\sqrt{7} - \sqrt{2}}$$= 9 + \sqrt{14}$
2	$\frac{2x^3 - 2x^2 - 24x - 7}{x^2 - x - 12}$ $= 2x + \left(\frac{-7}{x^2 - x - 12} \right)$ $= 2x + \left(\frac{-7}{(x-4)(x+3)} \right)$ $\frac{-7}{(x-4)(x+3)} = \frac{A}{x-4} + \frac{B}{x+3}$ $-7 = A(x+3) + B(x-4)$ $x = 4:$ $-7 = 7A$ $A = -1$ $x = -3;$ $-7 = -7B$ $B = 1$ $\frac{2x^3 - 2x^2 - 24x - 7}{x^2 - x - 12} = 2x - \frac{1}{x-4} + \frac{1}{x+3}$

3		$f(-2) = -8 + 4m - 12 - 4$ $4m - 24 = 0$ $m = 6 \text{ (shown)}$ (b) $\begin{array}{r} x^2 + 4x - 2 \\ (x+2)\sqrt{x^3 + 6x^2 + 6x - 4} \\ \underline{-x^3 + 2x^2} \\ 4x^2 + 6x \\ \underline{4x^2 + 8x} \\ -2x - 4 \\ \underline{-2x - 4} \\ f(x) = (x+2)(x^2 + 4x - 2) \\ x+2 = 0 \\ x = -2 \\ x^2 + 4x - 2 = 0 \\ x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ = \frac{-4 \pm \sqrt{4^2 - 4(1)(-2)}}{2(1)} \\ = \frac{-4 \pm \sqrt{24}}{2} \\ x = -2 \pm \sqrt{6} \end{array}$
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4	a Degree of numerator = 3, Degree of denominator = 2 Since degree of numerator > degree of denominator, $\frac{x^3 + 5x^2 - 14x - 10}{x^2 + 4x - 21}$ is not proper fraction b $ \begin{array}{r} x + 1 \\ \hline x^2 + 4x - 21 \overline{) x^3 + 5x^2 - 14x - 10} \\ \underline{-(x^3 + 4x^2 - 21x)} \\ x^2 + 7x - 10 \\ \underline{-(x^2 + 4x - 21)} \\ 3x + 11 \end{array} $ $\frac{x^3 + 5x^2 - 14x - 10}{x^2 + 4x - 21} = x + 1 + \frac{3x + 11}{x^2 + 4x - 21}$ Let $\frac{3x + 11}{(x + 7)(x - 3)} = \frac{A}{x + 7} + \frac{B}{x - 3}$ $3x + 11 = A(x - 3) + B(x + 7)$ When $x = -7$, $3(-7) + 11 = A(-7 - 3) + B(-7 + 7)$ $-10 = -10A$ $A = 1$ When $x = 3$, $3(3) + 11 = A(3 - 3) + B(3 + 7)$ $20 = 10B$ $B = 2$ Hence, $\frac{x^3 + 5x^2 - 14x - 10}{x^2 + 4x - 21} = x + 1 + \frac{1}{x + 7} + \frac{2}{x - 3}$.
5	a $2h(2)^3 + k(2)^2 + 4(2) + 4 = 4[2h(-1)^3 + k(-1)^2 + 4(-1) + 4]$ $16h + 4k + 12 = -8h + 4k$ $h = -\frac{1}{2}$ Since is a factor, $2\left(-\frac{1}{2}\right)(1)^3 + k(1)^2 + 4(1) + 4 = 0$ $k = -7$

	b $2hx^3 + kx^2 + 4x + 4 = 0$ $-x^3 - 7x^2 + 4x + 4 = 0$ $x^3 + 7x^2 - 4x - 4 = 0$ $x = 1 \left \begin{array}{rrrr} 1 & 7 & -4 & -4 \\ \downarrow & 1 & 8 & 4 \\ 1 & 8 & 4 & 0 \end{array} \right.$ $(x-1)(x^2 + 8x + 4) = 0$ $(x-1) = 0 \quad \text{or} \quad x^2 + 8x + 4 = 0$ $x^2 + 8x + 4 = 0$ $x = \frac{-8 \pm \sqrt{64-16}}{2}$ $= \frac{-8 \pm 4\sqrt{3}}{2}$ $= -4 \pm 2\sqrt{3}$ $\therefore x = 1, \quad -4 + 2\sqrt{3} \quad \text{or} \quad -4 - 2\sqrt{3}$
6	(a) $f(x) = 0$ $(x-1)(x-2)(x-k) = 0$ Since coefficient of $x^3 = -1$ $f(x) = -(x-1)(x-2)(x-k)$ $f(3) = 8$ $-(3-1)(3-2)(3-k) = 8$ $3-k = -4$ $k = 7$ (b) $f(-3) = -(-3-1)(-3-2)(-3-7)$ $= 200$